Week 11 Reading: Should we bias-correct climate models? (bring two questions Friday)

What bias correction assumes about a climate model, and what it can hide

2026-11-02

Read the assigned sections before Friday's class (11/6).

Assigned

1. Ehret et al. (2012) (open access), skipping §2.1-2.2.

Background

1. Sections 3.5.1-3.5.2 of Ghil et al. (2011).

Further reading

1. Lafferty & Sriver (2023) (open access), especially Fig. 6.
2. Sobel et al. (2023), Fig. 3 and the sections "The Time Frame" and "Solutions".

Bibliography

Ehret, U., Zehe, E., Wulfmeyer, V., Warrach-Sagi, K., & Liebert, J. (2012). Should We Apply Bias Correction to Global and Regional Climate Model Data?. *Hydrology and Earth System Sciences*, *16*(9), 3391–3404. https://doi.org/10.5194/hess-16-3391-2012

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv

Lafferty, D. C., & Sriver, R. L. (2023). Downscaling and Bias-Correction Contribute Considerable Uncertainty to Local Climate Projections in CMIP6. *Npj Climate and Atmospheric Science*, *6*(1), 1–13. https://doi.org/10.1038/s41612-023-00486-0

Sobel, A. H., Lee, C.-Y., Bowen, S. G., Camargo, S. J., Cane, M. A., Clement, A., et al. (2023). Near-Term Tropical Cyclone Risk and Coupled Earth System Model Biases. *Proceedings of the National Academy of Sciences*, *120*(33), e2209631120. https://doi.org/10.1073/pnas.2209631120